Sachin Ashok

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linkedin google scholar github

Research Interests

Networked Systems; with an emphasis on Microservices and Cloud Computing.

Education

2021 - **PhD.** in Computer Science
University of Illinois at Urbana-Champaign
Advisors: Dr. Radhika Mittal, Dr. P. Brighten Godfrey

2014 - 2018 **B.Tech.** in Computer Science and Engineering
National Institute of Technology, Trichy

Research Experience

2018 - 2020 **Microsoft Research, India** **Research Fellow**
Mentors: Dr. Venkat Padmanabhan, Dr. Johannes Gehrke

- **iBox: Internet In A Box**
  - Led the effort to design a data-informed network simulator for recreating the behaviour of paths in a target network such as the Internet.
  - Designed parameterised network models which included a cross-traffic model which was critical for recreating network behaviour in challenging settings like cellular networks.
  - Integrated iBox into NetEm and Skype's in-house simulator and made publicly available the derived iBoxNet profiles for network researchers.
- **R3Net: Reinforcement Learning for bandwidth control**
  - Coordinated on an effort across MSR India, MSR Asia & Microsoft Redmond to use reinforcement learning for bandwidth estimation & congestion control.
  - Developed realistic network environments to train RL agents.

Publications

- **Data-Driven Network Path Simulation with iBox**
  Sachin Ashok, Shubham Tiwari, Nagarajan Natarajan, Venkata N. Padmanabhan, Sundararajan Sellamanickam
  In Proceedings of ACM SIGMETRICS / IFIP Performance 2022.
- **Leveraging Service Meshes as a New Network Layer**
  Sachin Ashok, P. Brighten Godfrey, and Radhika Mittal
  In Proceedings of the 20th ACM Workshop on Hot Topics in Networks, HotNets '21.
- **iBox: Internet in a Box**
  Sachin Ashok, Sai Surya Duvvuri, Nagarajan Natarajan, Venkat Padmanabhan, Sundararajan Sellamanickam, Johannes Gehrke
  In Proceedings of the 19th ACM Workshop on Hot Topics in Networks, HotNets '20.
- **Reinforcement Learning for bandwidth estimation and congestion control in real-time communications**
  Joyce Fang, Martin Ellis, Bin Li, Siyao Liu, Yasaman Hosseinkashi, Michael Revow, Albert Sadovnikov, Ziyuan Liu, Peng Cheng, **Sachin Ashok**, David Zhao, Ross Cutler, Yan Lu, Johannes Gehrke
  MLForSystems, NeurIPS 2019.

Industry Experience

2016 **Mozilla Organization** **Intern**
Mentors: Dr. Franziskus Kiefer, Mr. Tim Taubert

◇ **Blake2 support for the Network Security Services library**

- Implemented a Blake2 (a cryptographic hash function) module for Mozilla's NSS library.
- The NSS library is used by Google, RedHat, and Mozilla products like Thunderbird.

2017 **Samsung R&D, India** **Intern**
Mentors: Dr. Pratibha Moogi, Mr. Karthikeyan Somanathan

◇ **Object localization**

- Developed an end to end system to leverage deep learning models using custom convolutional neural networks for localization of objects in a given image.
- Entire system pipeline (image capture to model inference) was pushed to production for Samsung mobile devices.

Honors

◇ Invited for demo at Microsoft TechFest, 2020.
◇ Among 45 students chosen worldwide for Mozilla Winter of Security.
◇ Among top 100 students chosen in India by Indian Institute of Science (IISc) for Summer School.
◇ Among top 25 students chosen in India for Research Programme by Institute of Mathematical Sciences (IMSc).
◇ Finalist, Microsoft Build The Shield, a national level contest held by Microsoft.
◇ Finalist, Pragyan CTF, an international level security contest held by NIT Trichy.

Talks

2020 iBox: Internet in a Box
HotNets 2020, University of Chicago, Virtual Event

2021 Leveraging Service Meshes as a New Network Layer
HotNets 2021, University of Cambridge, Virtual Event

2022 Leveraging Service Meshes as a New Network Layer
CSL Student Conference 2022, University of Illinois at Urbana-Champaign, Virtual Event

2022 Data-Driven Network Path Simulation with iBox
ACM SIGMETRICS'22, Indian Institute of Technology Bombay (IIT-B)

Skills

Languages: C++, C, Python
Frameworks: ns-2/3, Istio, Envoyproxy, Protobuf, MongoDB, gRPC